Kartik Karkera

+1 469 237 8194 | karkerakartik27@gmail.com | LinkedIn | GitHub | Portfolio

EXPERIENCE

Kapi Advisors Dallas, TX

Software Engineer Jul 2026 – Present

- Developed deterministic scoring microservices to normalize supplier quotations across currencies, pack sizes, discounts, taxes, and freight for consistent item-level comparisons.
- Built a Google OR-Tools CP-SAT optimization engine to minimize sourcing costs or maximize supplier scores while enforcing demand, budget, minimum-order, and supplier-count constraints.
- Integrated supplier quotes, pricing tiers, and eligibility through a data adapter, translating sourcing data into solver inputs and returning allocation recommendations for buyer review.
- Designed API and PostgreSQL models for buyer and supplier portals, supporting quotation submissions, clarification rounds, and approvals with auditable decision histories.

PyCray Technologies Noida, India

Data Engineer May 2023 – Jun 2024

- Engineered an event-driven pipeline using DynamoDB Streams and AppSync subscriptions to deliver real-time trading position updates with under 50 ms end-to-end latency.
- Designed a hot-cold architecture using DynamoDB for active trades and PostgreSQL for historical analytics, separating operational queries from reporting workloads.
- Built EventBridge-triggered Lambda ETL jobs to transfer trading records from DynamoDB to PostgreSQL, handling schema evolution and cross-store consistency.
- Implemented geospatial indexing in Elasticsearch to optimize location-based ride searches for a carpooling application.
- Developed trading analytics pipelines to aggregate active-user, open-position, and profit-and-loss metrics for daily business reporting.

PROJECTS

Traceline – Pipeline Anomaly Detection **1st Place, TidalHack @ TAMU**

Python, Flask, Pandas, NumPy, React

- Developed an ingestion and alignment pipeline for 15+ years of RCP Inc. inspection records, correlating scans to track corrosion growth across inspection runs.
- Designed a matching algorithm using fixed hardware anchors to correct odometer drift and align anomalies across inspection datasets.
- Built a React-based 3D dashboard to visualize pipeline anomalies and track their progression across inspections.

Data-Centric AI Challenge **3rd Place, 3LC x AWS Hackathon**

Python, ResNet-18, 3LC, AWS

- Achieved 94% validation accuracy on a ResNet-18 classifier through data quality improvements, without architecture changes, external data, or pretrained weights.
- Used the 3LC Dashboard to relabel low-confidence samples, rebalance classes, and adjust per-sample training weights based on embedding analysis.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java

Backend & APIs: FastAPI, Flask, REST APIs, GraphQL, Microservices

Cloud & DevOps: AWS (Lambda, EventBridge, DynamoDB Streams, AppSync), Docker, Linux, Git, GitHub Actions

Databases & Search: PostgreSQL, DynamoDB, MongoDB, Redis, Elasticsearch

AI & Data: Pandas, NumPy, RAG, LangGraph, Amazon Bedrock

Frontend & Testing: React, Tauri, Unit Testing, Integration Testing

EDUCATION

The University of Texas at Dallas Richardson, TX

Master of Science in Computer Science; GPA: 3.67/4.0 May 2026

CERTIFICATIONS

Databricks Certified Associate Developer for Apache Spark – Python Jan 2026

AWS Certified Data Engineer – Associate Sep 2025

AWS Certified Solutions Architect – Associate Jul 2025